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HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



ASSIGNMENT 2 DATASET PROPOSAL
TRASHCAN-MATERIAL SEMANTIC
SEGMENTATION

Dataset Selection and Evaluation Plan for Underwater Marine-Debris Segmentation

Course: CO3133 – Deep Learning and Its Applications

Semester: 261

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Class: CC01

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ASSIGNMENT 2 PROPOSAL OVERVIEW

The proposal was updated on 5 October 2026 with measured EDA and a video-disjoint split. Instructor approval is pending.

Item	Proposal details
Track	Semantic segmentation
A2 M1 deadline	7 October 2026, 23:59 (GMT+7)
Dataset decision	Use the TrashCan-Material taxonomy. Retain all 16 foreground categories and add background for 17 output classes. The eight trash categories include named materials, fishing gear and a miscellaneous category; ROV and biological labels retain their original IDs.

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1 Research Question and Task

Can a compact segmentation model with transferred knowledge identify underwater debris by material while distinguishing it from ROV equipment and marine life more accurately than a small U-Net trained from scratch? Within the pretrained model, does modest underwater color augmentation improve generalization to held-out videos without obscuring material cues?

Input: An RGB frame from an underwater remotely operated vehicle (ROV).

Output: A dense $H \times W$ class map. ID 0 is background; IDs 1–16 retain the exact dataset names and IDs listed in Table 1.1; ID 255 marks ambiguous overlap and letterbox padding and is excluded from loss and evaluation. Every pixel receives one semantic label. The archive contains 12,336 object annotations on 7,212 images.

Table 1.1: Exact category map and pooled annotation counts.

ID	Class name	Objects	Images	Videos
1	rov	3,317	2,721	145
2	plant	507	482	32
3	animal_fish	764	644	45
4	animal_starfish	398	134	15
5	animal_shells	249	139	7
6	animal_crab	309	186	15
7	animal_eel	343	259	35
8	animal_etc	235	221	25
9	trash_etc	2,039	1,740	153
10	trash_fabric	314	269	21
11	trash_fishing_gear	196	167	14
12	trash_metal	1,126	1,093	62
13	trash_paper	193	167	10
14	trash_plastic	1,864	1,753	97
15	trash_rubber	142	142	7
16	trash_wood	340	324	20

Image counts per class overlap because one image may contain several classes. Videos in this table count groups with nonignored semantic pixels. ID 0 is background; ID 255 is excluded, giving 17 predicted classes.

Use case: Visual scene understanding for marine-debris survey. This is a research benchmark, not an autonomous cleanup safety claim.

Dataset suitability: Local EDA verified 7,212 readable annotated images, comprising 6,008 original training images and 1,204 original validation images. There are 12,336 object annotations (9,741 training and 2,595 validation), 16 foreground classes and 312 video groups inferred from filenames. This exceeds the assignment scale requirement for segmentation. Published source-video counts describe the original collection; 312 is the count of filename groups in our acquired Material release.

2 Dataset Source and Preliminary Distribution

Working source: The group uses the [TrashCan-Material preliminary EDA folder](#) shared on Google Drive. It contains `dataset.zip`, Material COCO annotations, the preliminary notebook/report, QA figures and `LICENSE.txt`. The author source is the University of Minnesota IRV Lab and dataset repository; the imagery originates from JAMSTEC. The supplied local copy is identified by the hashes below. The original dataset, annotations and license were downloaded from the University of Minnesota’s official Data Repository.

Release/version: TrashCan 1.0, Material class configuration. The supplied `dataset.zip` is 553,029,970 bytes. SHA-256: `caf93d992b85343ccfc50038c94ca9ffa6c63e2284e04127753a40db8f33ba4c`. The selected files are `material_version/instances_train_trashcan.json` and `instances_val_trashcan.json`; their hashes are recorded in `metadata.json` and the split manifest. The archive includes both configurations, but this study uses Material only.

Annotation and conversion: The supplied COCO JSON files contain object polygon lists. Our versioned converter fills polygons on the native image canvas using Pillow 12.3.0 ImageDraw; coverage outside the canvas is clipped. Same-class masks are unioned. Cross-class overlapping pixels become ignore ID 255. Background is 0 and class IDs remain 1–16. The same helper will produce future training targets and evaluation ground truth. These derived semantic labels define our protocol; we will not compare them directly with published instance mask AP.

Data-use terms: The supplied `LICENSE.txt` permits academic teaching and research and requires attribution to the dataset authors (Jungseok Hong, Michael Fulton and Junaed

Sattar) and JAMSTEC; commercial use requires JAMSTEC permission. The COCO JSON license metadata separately records Attribution-NonCommercial-ShareAlike 2.0. We retain both records and the required attribution. The SegFormer model card links to the NVIDIA Source Code License, which limits use to noncommercial research or evaluation and requires retained license/attribution notices on redistribution.

Measured preliminary distribution: All 7,212 images contain annotations; none of the 12,336 annotations rasterizes to an empty mask under our converter. Images are 480×270 (3,967 images) or 480×360 (3,245 images). The original partitions have 306 and 133 filename video groups, with 127 groups shared between them. Class distribution is imbalanced: `trash_plastic` has 1,864 object annotations, while `trash_rubber` has 142 and occurs in only seven video groups. Figure 2.1 and Table 2.1 show measured counts and semantic pixel support. Counts use the full acquired dataset for preliminary characterization, before model selection.

Table 2.1: Pixel accounting on the original source partitions.

Source	Foreground (%)	Background (%)	Ignore (%)
Train	10.6016	89.3613	0.0371
Validation	10.6829	89.2572	0.0599

Percentages use the native image canvas and sum to 100%. Foreground union, including conflicting pixels, is 10.6387% for train and 10.7428% for validation. Supervised foreground excludes those conflicts. Pixel counts come from rasterized labels rather than summed annotation area.

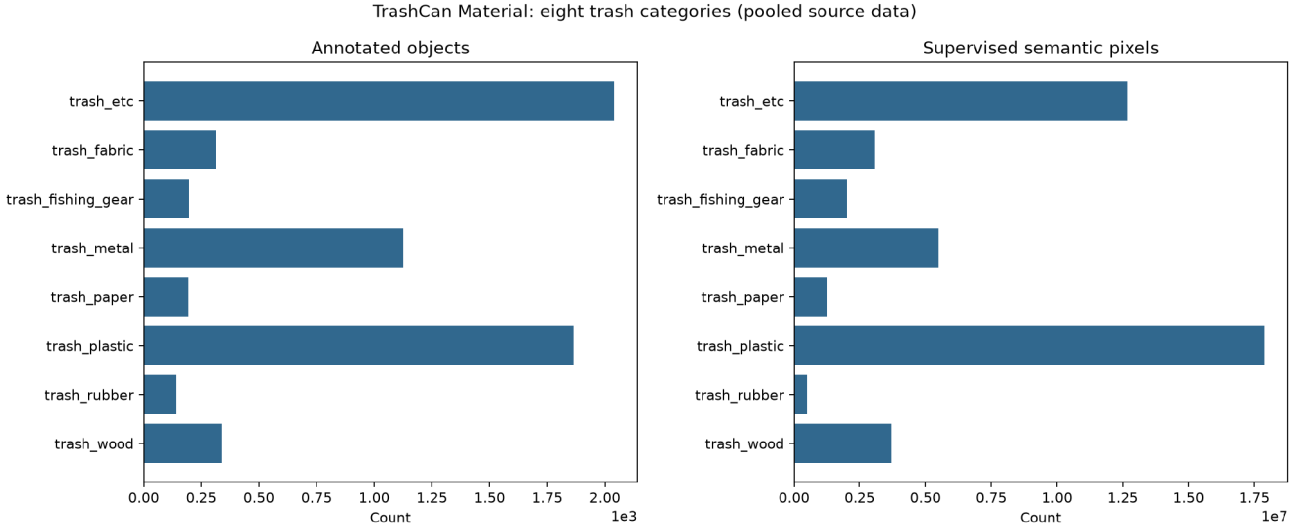


Figure 2.1: Pooled distribution of eight trash categories by objects and supervised semantic pixels. Source: group EDA of TrashCan 1.0; dataset authors and JAMSTEC.

3 Split and Leakage Prevention

Frozen split: We pooled the original train/validation metadata and assigned whole filename video groups with seed 36, targeting approximately 70/15/15 by image count. The resulting partitions contain 5,057 train, 1,077 validation and 1,078 test images, from 207, 54 and 51 videos. Every foreground class has at least three training videos and two videos in each held-out partition. Assignment balances image count and class support using metadata only, through 24 deterministic greedy/local-search candidates. No model predictions or scores were used. Image paths stay in their original folders; the manifest defines their new role.

Leakage evidence: We extract video groups from the `vid_*` prefix before `_frame` in each filename. The original split shares 127 groups and is therefore unsuitable for our video holdout. The new manifest has zero pairwise video overlap, covers every image exactly once, and records source paths and semantic-target hashes. SHA-256 checking of the supplied images found no exact duplicate files, but this does not establish independence of different videos from the same dive. Filename grouping is our available protection; deployment/domain generalization remains outside the claim. Test labels were used for initial class-support checks only; all subsequent model, loss, augmentation and stopping choices use training/validation.

Table 3.1: Video split and class coverage.

Split	Images	Videos	Annotations
Train	5,057	207	8,563
Validation	1,077	54	1,956
Test	1,078	51	1,817

Rare-class coverage: `trash_rubber` has 3/2/2 train/validation/test videos, `trash_paper` 6/2/2, and `trash_fishing_gear` 10/2/2. These checks guarantee presence, while the small number of held-out videos remains a limit on uncertainty and generalization claims.

Manifest: `eda_v1/splits/trashcan_material_seed36_v1.json`. Canonical content SHA-256: `87905f1a97dd2c04194e049ca61d1a7f3bbe2942e9fe17dc2d737c371cdce943`. The digest excludes the digest field itself; `verification.json` records the coverage and no-overlap checks.

4 Preprocessing and Evaluation Protocol

Label handling: Use the shared `rasterize_semantic` helper and the frozen category map to generate targets. There is no annotation-order overwrite priority: conflicting classes are ignored. Ignore pixels do not contribute to cross-entropy, Dice, mIoU or class weighting. Estimate any class weights or other learned preprocessing statistics using the new training partition only.

Image preparation: Preserve aspect ratio when letterboxing 480×270 or 480×360 images to 512×512 . RGB content uses bilinear interpolation; targets use nearest-neighbor. Black RGB padding receives target ID 255, so it does not contribute to loss or metrics. Apply matching geometric transforms to image and mask. Validation/test preprocessing is deterministic; normalization is fixed and documented for both systems.

Metrics: Primary metric is macro mIoU over the 16 foreground classes; macro foreground Dice is the companion metric. Report mIoU over trash IDs 9–16 separately, plus per-class IoU/Dice, pixel/video support and background-inclusive scores. Count only valid pixels; report ignore coverage. All 16 foreground classes are present in the frozen test split. For any future unsupported class, report its ground-truth and prediction support explicitly and use a stated evaluator convention instead of silently assigning a perfect score.

Qualitative analysis: Preliminary EDA includes six semantic overlays from six distinct videos, two per new partition, with an explicit file manifest so previous outputs cannot enter the report. Final evaluation will show preselected test scenes with RGB, ground

truth and model predictions. Analyze small/deformed debris, low visibility, confusion among material categories, ROV/biology confusion and boundaries. Include attribution to TrashCan authors and JAMSTEC on report figures.

Mask AP and bounding-box AP are not metrics for this proposal as they evaluate instance segmentation or detection and cannot be compared directly with semantic mIoU/Dice. Published TrashCan Mask R-CNN AP is context only, not a target score for our task.

5 Models and Controlled Experiment

Simple baseline: A four-stage U-Net with a lightweight convolutional encoder, initialized from scratch. Train with cross-entropy plus soft Dice loss, ignoring ID 255. Proposed initial settings are a 512×512 canvas, batch size 4, up to 40 epochs, and early-stopping patience 8. A post-approval throughput pilot will confirm feasibility before the main comparison; optimizer, learning-rate schedule and loss weights will then be frozen and recorded for both systems.

Modern/pretrained model: SegFormer-B0 initialized from [the SegFormer-B0 ADE 512 checkpoint](#), with the classifier replaced by a 17-class head and the complete model fine-tuned on TrashCan. The model card describes ImageNet-1k encoder pretraining followed by ADE20K segmentation fine-tuning. Our group’s masks/ignore convention will be used and logits are evaluated at the target resolution. The pretrained checkpoint has not yet been used for main experiments.

Comparison interpretation: U-Net versus SegFormer changes architecture and pretraining, so it is a system comparison rather than a causal isolation of transfer learning.

Controlled experiment: Compare SegFormer-B0 with color augmentation enabled versus disabled. On training RGB content, apply ColorJitter with probability 0.5: brightness, contrast and saturation factors are each sampled uniformly from $[0.9, 1.1]$, and hue shift from $[-0.01, 0.01]$; transformation order is randomized. Apply it before black letterbox padding and normalization; masks keep their labels. Validation/test RGB is not color-augmented. The augmentation bundle is the single changed factor. Keep split, initial weights, geometry, optimizer, schedule, loss, batch size, epoch budget, evaluator and run seed 69420 fixed.

Hypothesis: Modest appearance variation improves held-out-video robustness without damaging material cues. Retain augmentation if validation trash-only mIoU improves by at least 0.01 absolute and overall foreground mIoU and Dice each fall by no more than

0.01. These are prospective thresholds. Additional paired seeds 67 and 69 are optional compute-dependent repetitions, not completed runs.

Checkpoint selection: Select the highest validation foreground mIoU; break ties by lower validation loss and then the earlier epoch. Apply the same selection rule to every configuration. Early stopping uses validation only. After choices are locked, evaluate the selected baseline and pretrained configurations on the frozen test; do not revise configurations in response to test results.

6 Compute and Reproducibility

Compute and subset plan: Use all 7,212 labeled images and the frozen video split. Plan one 12–16 GB GPU, initially batch size 4 and a maximum of 40 epochs per run. Reserve approximately 12–24 GPU-hours for the three initial configurations at one common run seed; this is an estimate, not measured training time. Repeating at seeds 67 and 69 would roughly triple that budget and is conditional on pilot throughput and available compute. The pilot will measure memory, images/second, epoch time and storage before freezing the main budget. No subset is currently planned.

Reproducibility: Retain dataset/archive hashes, the converter version and Pillow version, split manifest/content hash, code commit, configuration, split seed and run seeds, hardware/software versions, train/validation curves, selected checkpoints, per-class metrics, training time, memory, parameter count and inference latency/throughput. The full archive, raw frames and large checkpoints remain outside the code repository. The group Drive folder holds working data/evidence; the public repository contains code, manifests, permitted figures and required attribution.

Sources

- TrashCan – [University of Minnesota IRV Lab](#).
- Hong, Fulton and Sattar, *TrashCan: A Semantically-Segmented Dataset towards Visual Detection of Marine Debris*, [paper](#).
- [University of Minnesota dataset record](#) and [repository record](#).
- [SegFormer-B0 pretrained checkpoint and model card](#).
- Dataset data-use record: supplied `LICENSE.txt` and COCO JSON licenses, retained in the accompanying evidence package.
- SegFormer model-use terms: [NVIDIA Source Code License for SegFormer](#).

- Measured evidence: `eda_v1/eda_report.md`, `metadata.json`, `split_summary.json` and the seed-36 manifest, included with this proposal.